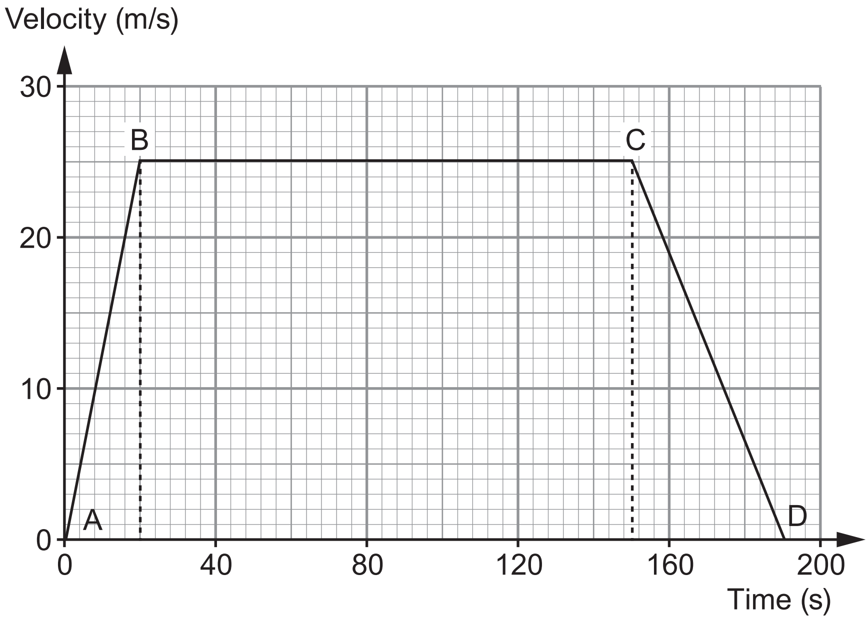
Examiner only

4. An empty school bus takes 190 seconds to travel between two sets of traffic lights. Its journey is shown on the velocity-time graph below.



(a) The bus accelerates uniformly from the first set of traffic lights at A. The resultant force acting on it is 5000 N. Use the graph and equations from page 2 to calculate the mass of the empty bus. [4]

Mass = ... kg

(b) The same resultant force acts on the bus which is now full of pupils. Explain how the start of the velocity-time graph (between A and B) would change. [2]

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Examiner only

(c) (i) State Newton's first law of motion. [2]

....

....

(ii) Explain how Newton's first law applies to part **B to C** of the journey. [2]

....

....

....

(d) During its journey from A to D how long is the bus moving at a velocity greater than 12.5 m/s? [2]

Time = ... s

(e) A lorry travels between the same two sets of traffic lights at a constant velocity without stopping. It takes the same time as the school bus. **On the graph opposite** draw the motion of the lorry. You may use equations from page 2 and the space below to show any calculations. [5]

Space for workings:

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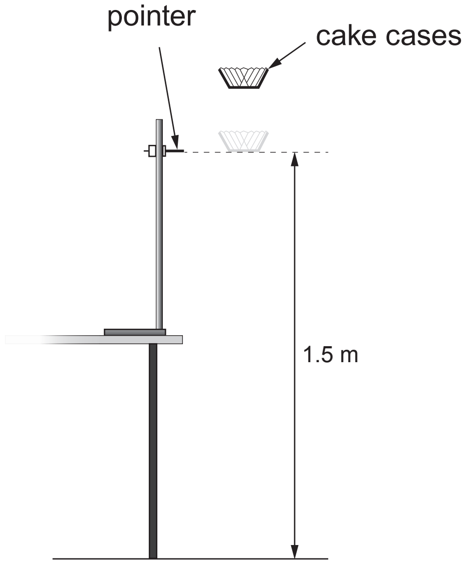
Examiner only

4. A group of students investigate how the terminal speed of falling paper cake cases depends on their mass. They use the apparatus shown in the diagram below.



The cake cases are dropped from at least 20 cm above the pointer. This allows them to reach a terminal speed by the time they reach the pointer. The time taken for the cases to fall the distance of 1.5 m between the pointer and the floor is measured.

The students are given a blank table to record their results. This is shown below.

Number of cake cases	Mass of cake cases (g)	Time taken for paper case cases to fall (s)						Terminal speed (m/s)
		Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Mean	
1								
2								
3								
4								
5								

(a) (i) The same cake cases are used throughout the experiment. State the other controlled **variables** in the experiment. [1]

....

....

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(ii) Explain the advantage of 5 trial timings for each number of cases. [2]

....

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(iii) The students are told the mean mass of a paper case. Explain how the quality of results collected is affected if the cases used each time are weighed rather than calculated by using the mean mass of a case. [2]

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(b) (i) Explain, in terms of forces, how the cake cases reach a terminal speed. [2]

....

....

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(ii) Describe how the experiment could be extended to check that the cases are falling with a terminal speed over the 1.5 m shown. [2]

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Examiner only

(c) The cake cases used are supplied in a box containing 250 cases. The combined mass of the box **and** cases is 118 g. The mass of the empty box is 18 g. The terminal speed of 4 paper cases is found to be 2.54 m/s.
The weight of a 1 kg mass on Earth is 10 N.

Use equations from page 2 to calculate:

(i) the time taken for the 4 paper cases to fall the 1.5 m shown in the diagram on page 9. [2]

Time = ... s

(ii) the air resistance acting on 4 paper cases when they fall at terminal speed.
Show your working. [4]

Air resistance = ... N

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Examiner only

2. A group of students investigate how the surface area of a falling paper cake case affects its terminal speed.

- Cake case 1 has a mass of 0.5 g and a surface area of 100 cm^2.
- Cake case 1 is dropped from a height of 1.80 m but only timed over the final 1.50 m of the fall.

The students' results are shown in the table below.

Drop time (s)			Mean drop time (s)	Drop distance (m)
Attempt 1	Attempt 2	Attempt 3		
0.96	0.92	0.94	....	1.50

(a) (i) The students decide there are no anomalies. Explain why. [1]

....

(ii) **Complete the table** to show the mean drop time. [1]
Space for calculation.

(b) The experiment is repeated with cake case 2.
It has the same shape and the same mass as cake case 1.
However, cake case 2 has a surface area of 50 cm^2.
The students correctly calculate the terminal speed for both cake cases.

Cake case 1		
Mass (g)	Surface area (cm^2)	Terminal speed (m/s)
0.5	100	1.6

Cake case 2		
Mass (g)	Surface area (cm^2)	Terminal speed (m/s)
0.5	50	2.3

Examiner only

(i) A cake case reaches terminal speed when its weight is balanced by air resistance. **Tick (✓)** the **three** correct statements. [3]

Cake case 2 has the same terminal speed as cake case 1.	☐
Cake cases 1 and 2 have identical weight.	☐
At terminal speed, cake case 1 experiences a greater value of air resistance than cake case 2.	☐
At terminal speed, both cake cases experience identical values of air resistance.	☐
At terminal speed, cake case 1 experiences a smaller value of air resistance than cake case 2.	☐
At terminal speed, both cake cases have zero acceleration.	☐

(ii) Before the experiment was carried out the students made the following prediction:

"If the surface area of the cake case is halved its terminal speed will double."

Use data from the tables on the previous page to explain whether their prediction was correct. [2]

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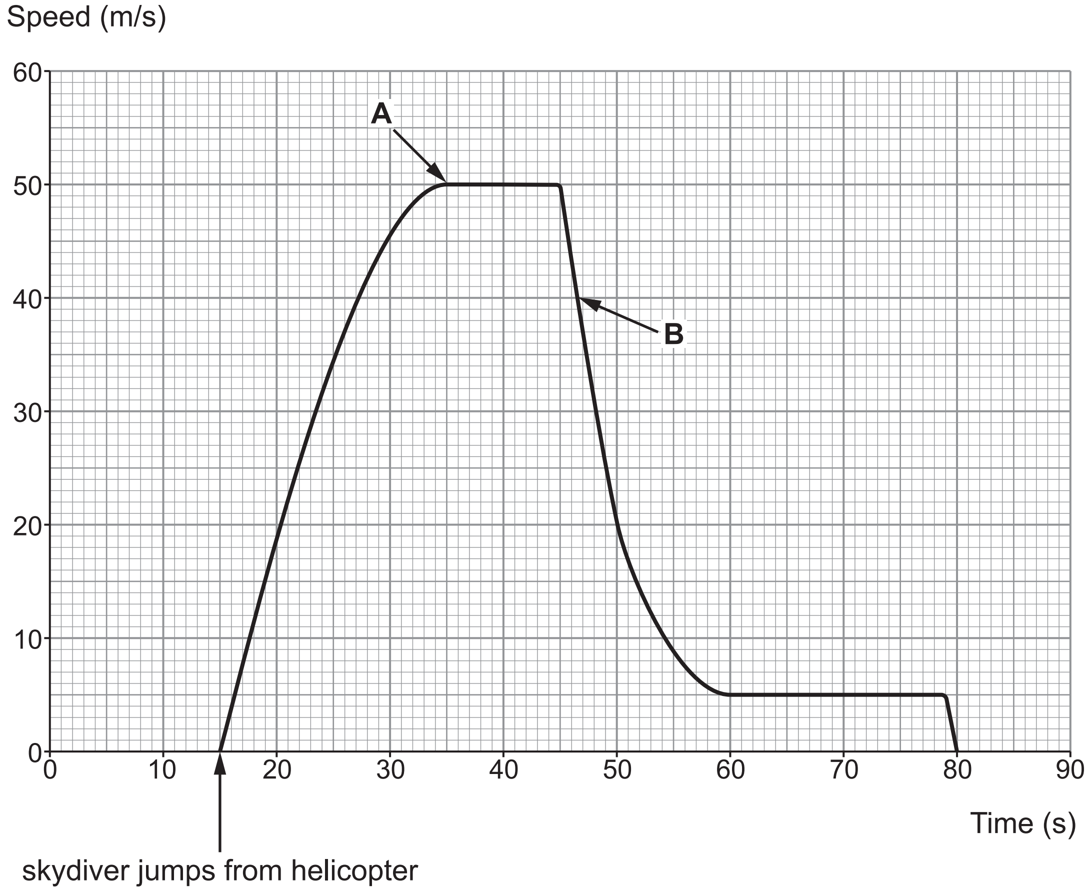
Examiner only

(c) A skydiver sits at the doorway of a helicopter for 15 s before jumping. His speed is recorded and displayed on the graph below.



Tick (✓) the **three** correct statements. [3]

The skydiver lands on the ground 80s after jumping from the helicopter.	☐
The terminal speed after the parachute is opened is $\frac{1}{10}$th of the terminal speed before the parachute is opened.	☐
The skydiver's weight is greatest at the point labelled **B** on the graph.	☐
The parachute is opened 30 s after the skydiver leaves the helicopter.	☐
At point **B** the weight of the skydiver is greater than the air resistance.	☐
At point **A** the skydiver stops accelerating.	☐

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Examiner only

7. (a) State Newton's **three** laws of motion **and** explain how they each apply to the motion of skydivers from when they first jump from a plane until they first reach terminal speed. [6 QER]

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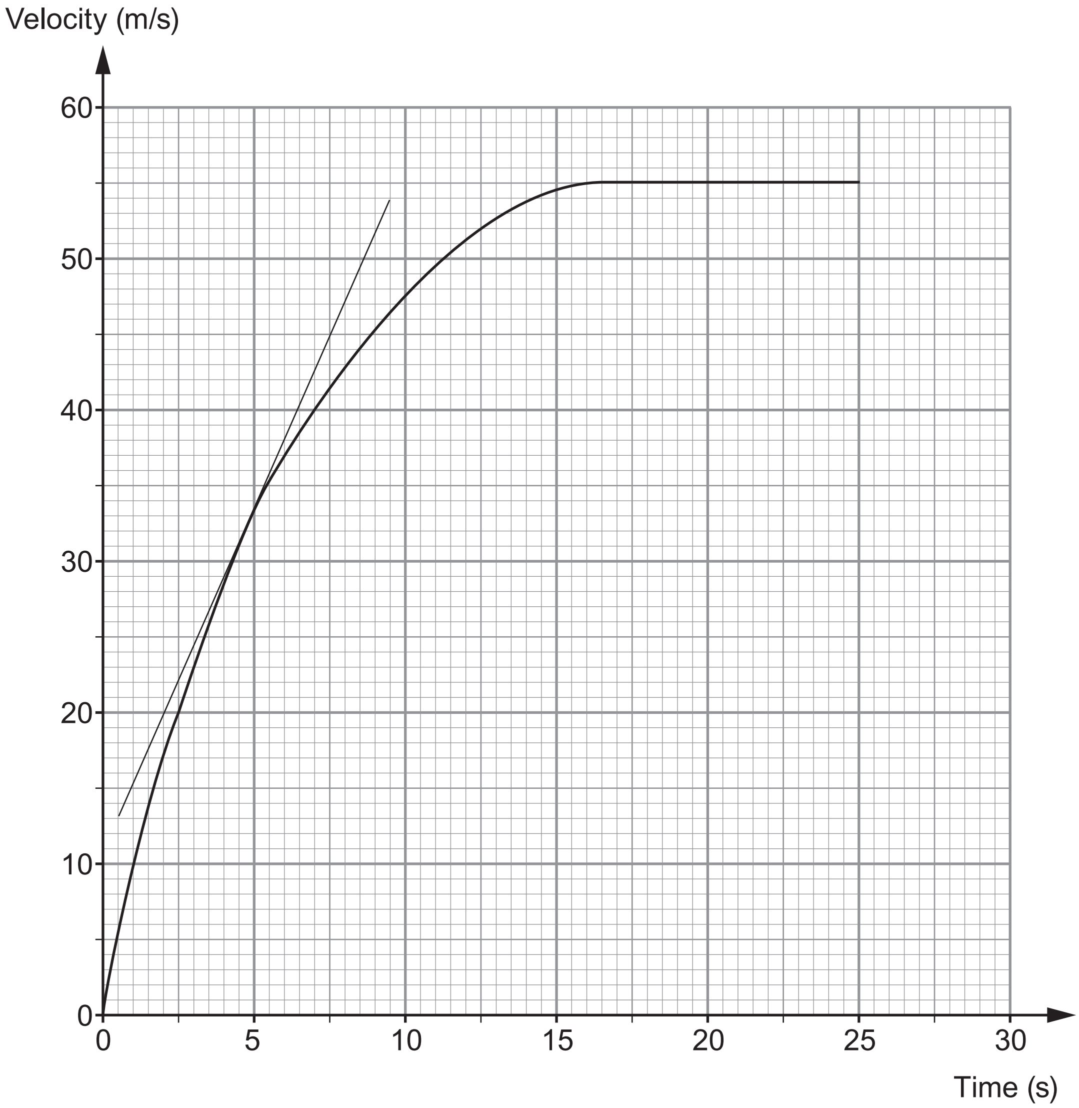
(b) The velocity-time graph below shows the motion of a skydiver.
A tangent has been drawn at 5 seconds.



Examiner only

(i) Acceleration can be calculated by measuring the gradient of a velocity-time graph. Calculate the acceleration of the skydiver at **5 s** by using the tangent shown. [3]

Acceleration = ... m/s^2

(ii) Describe how the acceleration changes over the 25 s shown. [2]

....

....

....

(iii) Use the graph and an equation from page 2 to estimate the distance travelled by the skydiver in the **first 5 s**. [3]

Distance travelled = ... m

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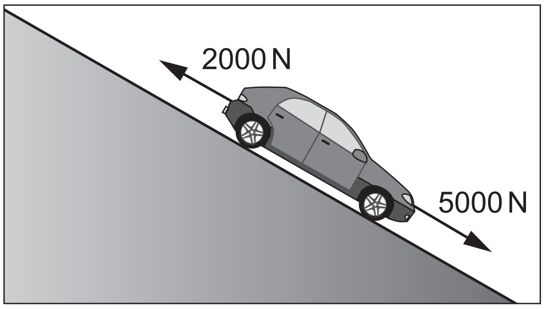
Examiner only

2. The diagram shows a car rolling down a slope. Two of the forces acting on the car are labelled.



(a) The car has a **weight** of 10 000 N. Use the equation:

$$\text{mass} = \frac{\text{weight}}{\text{gravitational field strength}}$$

to calculate the mass of the car. [2]
(Gravitational field strength, g, = 10 N/kg)

mass = ... kg

(b) Use the information in the diagram to answer the following questions.

(i) Calculate the resultant force acting down the slope. [2]

resultant force = ... N

(ii) Use your answers from parts (a) and (b)(i) and the equation:

$$\text{acceleration} = \frac{\text{resultant force}}{\text{mass}}$$

to calculate the acceleration of the car at the instant shown and state the unit. [3]

acceleration = ...

unit = ...

Examiner only

(iii) I. Explain how the resultant force on the car changes as it speeds up. [2]

....

....

....

II. State how this change in resultant force affects the acceleration of the car. [1]

....

(c) At the bottom of the slope the car continues horizontally at a constant speed of 12 m/s with a kinetic energy of 72 000 J.

(i) State **one** reason why the potential energy at the top of the hill must have been greater than 72 000 J. [1]

....

(ii) At the bottom of the hill a braking force is applied which stops the car over a distance of 15 m.

Use the equation:

$$\text{force} = \frac{\text{work done}}{\text{distance}}$$

to calculate the braking force. [2]

braking force = ... N

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Examiner only

8. Newton's laws describe the motion of objects.

(a) (i) State what quantity is an expression of the inertia of a body. [1]

....

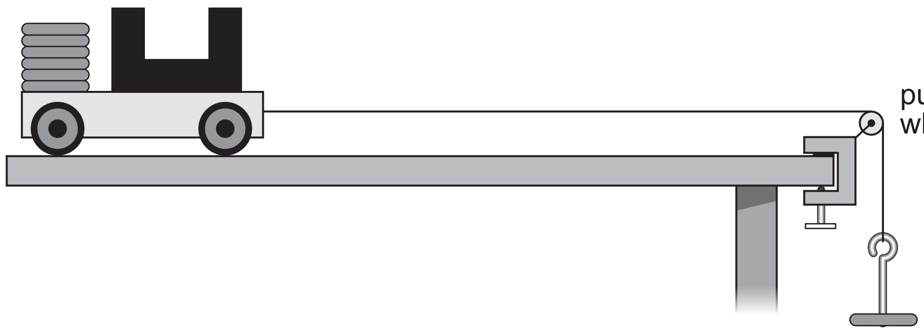
(ii) State Newton's 1st law. [2]

....

....

....

(b) Students investigate Newton's 2nd law.
They measure the acceleration of a trolley for different forces using light gates and a datalogger.
The method they follow is given below.



Method

1. Attach a 50 g mass hanger to the string. This gives an applied force of 0.5 N.
2. Release the trolley and record the acceleration from the datalogger.
3. Repeat step 2 twice more to give 3 results.
4. Remove one of the 50 g slotted masses from the trolley and place it on the mass hanger to increase the applied force by 0.5 N.
5. Repeat steps 2 to 4 until all the slotted masses have been placed on the mass hanger.

(i) State the independent variable. [1]

....

(ii) State the dependent variable. [1]

....

Examiner only

(iii) The same trolley is used throughout the experiment.
State **one** other controlled variable. [1]

....

(iv) Some of the students' results are shown below.

Applied force (N)	Acceleration (m/s^2)			
	Trial 1	Trial 2	Trial 3	Mean
0.5	0.299	0.323	0.309	0.310
1.0	0.618	0.619	0.621	0.619
1.5	0.923	0.936	0.934	0.931

Sara states that the results prove Newton's 2nd law because the acceleration is proportional to the applied force.
Explain whether you agree. [2]

....

....

....

(v) I. Use the results for an applied force of 0.5 N and an equation from page 2 to calculate the total mass of the trolley and the slotted masses. [3]

mass = ... kg

II. Explain how the students could check the accuracy of this value of the mass. [1]

....

....

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END OF PAPER